

Exercises 8.A

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Exercise 1. Define $T \in \mathcal{L}(\mathbb{C}^2)$ by

$$T(w, z) = (z, 0).$$

Find all generalized eigenvectors of T .

Proof. We begin by finding the eigenvalue(s) of T . We have $\lambda(w, z) = (z, 0)$, so

$$\begin{aligned}\lambda w &= z \\ \lambda z &= 0.\end{aligned}$$

Thus either $\lambda = 0$ or $z = w = 0$. The latter case yields the zero vector, so λ is the only eigenvalue. Since $\mathbb{C}^2$ has dimension 2, we find

$$\begin{aligned}(T - \lambda I)^2(w, z) &= T^2(w, z) \\ &= (0, 0),\end{aligned}$$

and conclude that every $v \in V$ is a generalized eigenvector of T corresponding to 0. $\square$

Exercise 2. Define $T \in \mathcal{L}(\mathbb{C}^2)$ by

$$T(w, z) = (-z, w).$$

Find the generalized eigenspaces corresponding to the distinct eigenvalues of T .

Proof. First we find the eigenvalues of T . We get

$$\lambda w = -z \tag{1}$$

$$\lambda z = w. \tag{2}$$

Taking $z \neq 0$ and $\lambda \neq 0$ (either would produce $z = w = 0$, the zero vector), then (1) yields $w = -\frac{z}{\lambda}$. Combining (2) with (1), we see that

$$\begin{aligned}\lambda z = -\frac{z}{\lambda} &\implies \lambda^2 z = -z \\ &\implies \lambda^2 = -1.\end{aligned}$$

Thus the eigenvalues of T are i and $-i$.

Observe that both $E(i, T)$ and $E(-i, T)$ have dimension at least 1, and since their nonzero elements must be linearly independent, given $v_1 \in E(i, T)$, $v_2 \in E(-i, T)$ with $v_1, v_2 \neq 0$, v_1, v_2 spans V . Since $E(i, T) \subset G(i, T)$ and $E(-i, T) \subset G(-i, T)$, $v_1 \in G(i, T)$ and $v_2 \in G(-i, T)$, which together form a basis of V . Thus $G(i, T)$ and $G(-i, T)$ each have dimension 1, so $E(i, T) = G(i, T)$ and $E(-i, T) = G(-i, T)$. It will suffice, then, to find the eigenspaces.

Taking equation (2) from above and $\lambda = i$, we get that $iz = w$, so

$$E(i, T) = G(i, T) = \{(iz, z) | z \in \mathbb{C}\}$$

and likewise taking $\lambda = -i$ we get

$$E(-i, T) = G(-i, T) = \{(-iz, z) | z \in \mathbb{C}\}.$$

□

Exercise 3. Suppose $T \in \mathcal{L}(V)$ is invertible. Prove that $G(\lambda, T) = G(\frac{1}{\lambda}, T^{-1})$ for every $\lambda \in \mathbf{F}$ with $\lambda \neq 0$.

Proof. Given $\lambda \in \mathbf{F}$ with $\lambda \neq 0$, let $v \in G(\lambda, T)$. Then $(T - \lambda I)^{\dim V} v = 0$ by definition. Observe that

$$\begin{aligned} (T^{-1} - \frac{1}{\lambda} I)^{\dim V} (T - \lambda I)^{\dim V} v &= ((T^{-1} - \frac{1}{\lambda} I)(T - \lambda I))^{\dim V} v \\ &= (I - \lambda T^{-1} - \frac{1}{\lambda} T + I)^{\dim V} v \\ &= (2I - \lambda T^{-1} - \frac{1}{\lambda} T)^{\dim V} v \\ &= (2I - \lambda T^{-1} - \frac{1}{\lambda} T)^2 (2I - \lambda T^{-1} - \frac{1}{\lambda} T)^{\dim V - 2} v \\ &= (4I - 2\lambda T)^2 (2I - \lambda T^{-1} - \frac{1}{\lambda} T)^{\dim V - 2} v \end{aligned}$$

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□

Exercise 4. Suppose $T \in \mathcal{L}(V)$ and $\alpha, \beta \in \mathbf{F}$ with $\alpha \neq \beta$. Prove

$$G(\alpha, T) \cap G(\beta, T) = \{0\}.$$

Proof. Suppose, to the contrary, that $v \in G(\alpha, T)$ and $v \in G(\beta, T)$ for some nonzero $v \in V$. Then by 8.13, the list v, v is linearly independent, which is absurd. □

Exercise 5. Suppose $T \in \mathcal{L}(V)$, m is a positive integer, and $v \in V$ is such that $T^{m-1}v \neq 0$ but $T^m v = 0$. Prove that

$$v, Tv, T^2v, \dots, T^{m-1}v$$

is linearly independent.

Proof. First, note that $T^j v \neq 0 \forall j \in \mathbb{Z}$ such that $0 \leq j \leq m-1$ since $T^{m-1} v \neq 0$. We now seek a contradiction by induction. Suppose that $Tv = c_0 v$ for some $c_0 \in \mathbf{F}$. Then $c_0 \neq 0$ and

$$T^2 v = T(Tv) = T(c_0 v) = c_0 Tv$$

so since $c_0 \neq 0$ and $Tv \neq 0$, $T^2 v \neq 0$. Now assume that for some $j \in \mathbb{Z}$ such that $1 \leq j \leq m-1$, the list $v, Tv, \dots, T^{j-1} v$ is linearly independent. Assume further, to contradict our desired result, that the list $v, Tv, \dots, T^j v$ is linearly dependent. Then there exist $c_0, \dots, c_j \in \mathbf{F}$ not all zero such that

$$0 = c_0 v + c_1 Tv + \dots + c_j T^j v.$$

Then

$$-c_j T^j v = c_0 v + \dots + c_{j-1} T^{j-1} v.$$

If $c_j = 0$, then

$$0 = c_0 v + \dots + c_{j-1} T^{j-1} v,$$

so $c_0 = \dots = c_{j-1} = 0$ by the induction hypothesis, contradicting our claim that $c_0, \dots, c_j$ are not all zero. Thus $c_j \neq 0$ and we can define $d_k = -c_k/c_j$ for k an integer where $1 \leq k \leq j-1$, by which we get

$$T^j v = d_0 v + \dots + d_{j-1} T^{j-1} v,$$

and $d_0, \dots, d_{j-1}$ are not all zero since $T^j v \neq 0$.

Now we can consider $T^{j+1} v$ with the aim of showing that it does not send v to 0, which would contradict the case of $j = m-1$, thus completing the proof. We have

$$\begin{aligned} T^{j+1} v &= T(T^j v) \\ &= T(d_0 v + \dots + d_{j-2} T^{j-2} v + d_{j-1} T^{j-1} v) \\ &= d_0 Tv + \dots + d_{j-2} T^{j-1} v + d_{j-1} T^j v \\ &= d_0 Tv + \dots + d_{j-2} T^{j-1} v + d_{j-1} (d_0 v + \dots + d_{j-1} T^{j-1} v) \\ &= d_0 Tv + \dots + d_{j-2} T^{j-1} v + d_{j-1} d_0 v + \dots + d_{j-1}^2 T^{j-1} v \\ &= d_{j-1} d_0 v + (d_0 + d_{j-1} d_1) Tv + \dots + (d_{j-2} + d_{j-1}^2) T^{j-1} v. \end{aligned}$$

If $T^{j+1} v$ is equal to zero, then by the linear independence of $v, Tv, \dots, T^{j-1} v$, we have

$$0 = d_{j-1} d_0 = d_0 + d_{j-1} d_1 = \dots = d_{j-2} + d_{j-1}^2. \quad (3)$$

While most of these equalities appear a bit cumbersome, the first equality provides us with two nice cases to deal with. Namely, either $d_{j-1} = 0$ or $d_0 = 0$. Taking the former, (3) becomes

$$0 = 0 = d_0 = d_1 = \dots = d_{j-2},$$

which contradicts $d_0, \dots, d_{j-1}$ not all being zero. On the other hand, take $d_0 = 0$ and (3) becomes

$$0 = 0 = d_{j-1}d_1 = d_1 + d_{j-1}d_2 = \dots = d_{j-2} + d_{j-1}^2.$$

This brings us to one final consideration; we have already shown that $d_{j-1} \neq 0$, so we must have $d_1 = 0$, which makes the fourth equality $0 = d_{j-1}d_2$, implying that $d_2 = 0$. Continuing on in this manner, we see that once again $0 = d_0 = d_1 = \dots = d_{j-1}$, yielding the same contradiction as before. $\square$

Exercise 6. Suppose $T \in \mathcal{L}(\mathbb{C}^3)$ is defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that T has no square root. More precisely, prove that there does not exist $S \in \mathcal{L}(\mathbb{C}^3)$ such that $S^2 = T$.

Proof. If such an S exists, then since T is nilpotent (T^3 is the zero map), S is also nilpotent ($S^6 = T^3$). But that would mean that S^3 is the zero map by 8.18, which contradicts the fact that $S^4 = T^2$, and $T^2(z_1, z_2, z_3) = (z_3, 0, 0)$, which is not the zero map. $\square$

Exercise 7. Suppose $N \in \mathcal{L}(V)$ is nilpotent. Prove that 0 is the only eigenvalue of N .

Proof. First we show that 0 is in fact an eigenvalue of N . If it is not, then $\text{null } N = \{0\}$. Take $v \in V$. Then $Nv \neq 0$, and in fact $N^k v \neq 0 \forall k \in \mathbb{Z}^+$, contradicting the fact that N is nilpotent.

Now assume that $\lambda \in \mathbf{F}$ is some nonzero eigenvalue of N . Then if v is an eigenvector of N corresponding to λ , $N^k v = \lambda^k v \neq 0$, contradicting nilpotency. $\square$

Exercise 8. Prove or give a counterexample: The set of nilpotent operators on V is a subspace of $\mathcal{L}(V)$.

Proof. Consider $T, S \in \mathcal{L}(C^2)$ defined by $T(v_1, v_2) = (v_2, 0)$ and $S(v_1, v_2) = (0, v_1)$. Both of these are nilpotent, but $(T + S)(v_1, v_2) = (v_2, v_1)$. If $T + S$ were nilpotent, $(T + S)^2$ would be the zero map since $\dim V = 2$, but in fact $(T + S)^2(v_1, v_2) = (v_1, v_2)$, which is, of course, not the zero map. $\square$

Exercise 9. Suppose $S, T \in \mathcal{L}(V)$ and ST is nilpotent. Prove that TS is nilpotent.

Proof. Of course, $(ST)^{\dim V} = 0$ since ST is nilpotent. Given $v \in V$, we have

$$\begin{aligned} (TS)^{\dim V + 1}v &= T(ST^{\dim V})Sv \\ &= T(0)Sv \\ &= T((0)(Sv)) \\ &= T(0) \\ &= 0 \end{aligned}$$

where " (0) " here is the zero map. Thus $(TS)^{\dim V + 1} = (0)$, so TS is nilpotent. $\square$